

Chapter 03 – Conceptualizing Interaction

Summary

1. Introduction to Conceptualizing Interaction

- **Definition:** Conceptualizing interaction involves creating a high-level framework (a *conceptual model*) to define how users will interact with a system, what it will do, and how it aligns with user needs and goals.
- **Purpose:**
 - Scrutinize vague ideas to assess feasibility, desirability, and usefulness.
 - Reduce risks in early design stages by addressing unknowns (e.g., user acceptance, technical constraints).

2. Key Concepts

A. Conceptual Model

- **Definition:** A high-level description of a system's organization and operation (Johnson & Henderson, 2002).
 - **Components:**
 1. **Metaphors/Analogies:** Leverage familiar concepts (e.g., "desktop" for file organization).
 2. **Task-Domain Objects:** Elements users interact with (e.g., "files," "folders").
 3. **Operations:** Actions users perform (e.g., "save," "delete").
 - **Example:** The Xerox Star (1981) used an office metaphor (icons for files, folders) to make digital interactions intuitive.

B. Assumptions vs. Claims

- **Assumptions:** Untested beliefs (e.g., "Users will enjoy conversing with a robot waiter").
- **Claims:** Assertions needing validation (e.g., "The robot will increase restaurant revenue").

- **Critical Analysis:** Challenge assumptions/claims to avoid unrealistic designs (e.g., 3D TV's failure due to overstated user interest).

C. Problem Space Framework

- Ask:
 - What problems exist with current systems?
 - How will the new design solve them?
 - Does it extend or replace existing workflows?

3. Interface Metaphors

- **Purpose:** Bridge familiarity (real-world knowledge) with new digital interactions.
 - **Examples:**
 - **Shopping Cart:** Mimics physical shopping for e-commerce.
 - **Card Metaphor:** Fast-food kiosks use "cards" for items, aiding selection.
 - **Benefits:** Accelerates learning, enhances accessibility.

4. Interaction Types

Five core types guide how users engage with systems:

Type	Description	Examples
Instructing	Command-based interactions (efficient for repetitive tasks).	Vending machines, CLI tools.
Conversing	Dialogue-like (voice/text).	Chatbots (Siri), phone menus.
Manipulating	Direct control of virtual objects (drag, zoom).	Photoshop, AR object placement.
Exploring	Navigating virtual/physical spaces.	Google Maps, VR data visualization.

Type	Description	Examples
Responding	System-initiated alerts (e.g., fitness milestones).	Smartwatch notifications, Google Lens.

- **Trade-offs:**
 - *Conversing*: Natural but prone to misunderstandings.
 - *Direct Manipulation*: Intuitive but complex to design (e.g., Shneiderman's principles: visibility, feedback).

5. Design Principles & Challenges

- **Systems View**: Consider interdependencies (e.g., a robot waiter's impact on staff dynamics).
- **Metaphor Pitfalls**: Over-reliance can limit innovation (e.g., skeuomorphic design).
- **Hybrid Models**: Combine interaction types (e.g., voice + touch for smart TVs).

6. Key Takeaways

1. **Conceptual Models** ground design in user mental models and task needs.
2. **Metaphors** simplify onboarding but require cultural/contextual fit.
3. **Interaction Types** shape UX: choose based on task nature (e.g., *exploring* for VR, *instructing* for workflows).
4. **Validation** is critical: Test assumptions early (e.g., will users trust a robot server?).

Quote: "*The best conceptual models appear obvious and simple.*" (Slide 23)

Example Analysis:

- **Robot Waiter vs. Functional Server:**
 - *Speaking Robot*: Assumes entertainment value but risks

annoyance.

- *Tray-Based Server*: Solves a clear problem (labor shortage) with minimal friction.